

RESTAssured и тест контроллера. Использовала Testcontainers.

The screenshot displays an IDE with a project structure on the left and a code editor on the right. The project structure includes folders for `model`, `repository`, `service`, `utils`, `CoursesApplication`, `resources`, `test`, `ru.natali.courses`, `controller`, `it`, `service`, `resources`, and `screens`. The `LoginControllerTest` file is selected in the `controller` folder.

The code editor shows the following Java code:

```
33 static void postgresProperties(DynamicPropertyRegistry registry) {
34     registry.add(name: "spring.datasource.username", postgresSQLContainer::getUsername);
35     registry.add(name: "spring.datasource.password", postgresSQLContainer::getPassword);
36 }
37
38
39 @BeforeEach
40 void setup() {
41     // Настраиваем RESTAssured на работу с MockMvc
42     RestAssuredMockMvc.webApplicationContextSetup(webApplicationContext);
43 }
44
45 @Test
46 void testLoginPage() {
47     given() MockMvcRequestSpecification
48         .when() MockMvcRequestAsyncSender
49         .get(s: "/login") MockMvcResponse
50         .then() ValidatableMockMvcResponse
51         .statusCode(i: 200)
52         .body(containsString(substring: "login")); // Проверяем, что возвращается страница login
53 }
54 }
```

The bottom of the screenshot shows the test execution results. The test `LoginControllerTest.testLoginPage` passed successfully, taking 826 ms. The output log shows the following messages:

```
17:45:32,631 |-INFO in ch.qos.logback.classic.LoggerContext[default] - This is logback-classic version 1.5.18
17:45:32,634 |-INFO in ch.qos.logback.classic.util.ContextInitializer@6ff29830 - Here is a list of configurators disc
17:45:32,634 |-INFO in ch.qos.logback.classic.util.ContextInitializer@6ff29830 - org.springframework.boot.logging.l
17:45:32,634 |-INFO in ch.qos.logback.classic.util.ContextInitializer@6ff29830 - They will be invoked in order until
17:45:32,634 |-INFO in ch.qos.logback.classic.util.ContextInitializer@6ff29830 - Constructed configurator of type cla
```

Courses F:\Innopromjects\Hamcrest\DZHamcrest\Ha

> .idea

> .mvn

> logs

app.log

> src

> main

> java

> ru.natali.courses

> aop

> config

> controller

> dto

> model

> repository

> service

> utils

CoursesApplication

> resources

> test

> java

> ru.natali.courses

> controller

LoginControllerTest

> it

CourseServiceIntegrationTest

> service

CourseArchivingServiceTest

21 class CourseServiceIntegrationTest {

39 @Test

40 void shouldSaveAndRetrieveCourse() {

41 Course course = new Course();

42 course.setName("Integration Test");

43 course.setStartDate(LocalDate.now()); // ← Устанавливаем обязательное поле

44 course.setActive(true);

45

46 Course saved = courseService.createCourse(course);

47 Course found = courseService.getCourseById(saved.getId());

48

49 assertThat(found.getName(), is(value: "Integration Test"));

50 assertThat(found.getStartDate(), is(course.getStartDate()));

51 }

52

53 @Test

54 void testCourseName() {

55 Course course = new Course();

56 course.setName("Spring Boot");

57

58 assertThat(course.getName(),

59 allOf(

60 is(value: "Spring Boot"),

61 containsString(substring: "Boot"),

62 not(emptyString()),

63 notNullValue()

64));

65 }

CourseServiceIntegrationTest

CourseServiceIntegrationTest

Tests passed: 2 of 2 tests - 381 ms

shouldSaveAndRetrie 374 ms

testCourseName() 7 ms

C:\Users\natal\jdk\jbr-17.0.12-1\bin\java.exe ...

17:54:39,027 |-INFO in ch.qos.logback.classic.LoggerContext[default] - This is logback-classic version 1.5.18

17:54:39,029 |-INFO in ch.qos.logback.classic.util.ContextInitializer@23941fb4 - Here is a list of configurators discovered as a service, by rank:

17:54:39,029 |-INFO in ch.qos.logback.classic.util.ContextInitializer@23941fb4 - org.springframework.boot.logging.logback

Project

> dto

> model

> repository

> service

> utils

CoursesApplication

> resources

test

java

ru.natali.courses

controller

LoginControllerTest

LoginControllerTest.java

CourseServiceTest.java

CourseServiceIntegrationTest.java

28

29

30

31

32

33

34

35

36

37

38

39

import static org.mockito.Mockito.when;

import static org.hamcrest.MatcherAssert.assertThat;

import static org.hamcrest.Matchers.*;

@ExtendWith(MockitoExtension.class)

class CourseServiceTest {

@Mock 18 usages

private CourseRepository courseRepository;

@Mock 6 usages

run

CourseServiceTest

✓ CourseServiceTest (ru.natali.courses: 728 ms)

✓ Tests passed: 23 of 23 tests – 728 ms

✓ updateCourse_ShouldUpdateCourse: 666 ms

✓ getArchivedCourses_ShouldReturnOnlyArchivedCourses: 2 ms

✓ getArchivedCourses_ShouldReturnEmptyList: 2 ms

✓ getAllCourses_ShouldReturnAllCourses: 15 ms

✓ getCourseStatisticsReport_ShouldReturnStatisticsReport: 2 ms

✓ updateCourse_ShouldNotChangeUnpublishedCourses: 2 ms

✓ getCoursesForStudent_ShouldCorrectlyFilterCourses: 2 ms

✓ deleteCourse_ShouldThrowExceptionIfCourseDoesNotExist: 2 ms

✓ deleteCourse_ShouldDeactivateCourse: 1 ms

✓ getCourseById_ShouldThrowExceptionIfCourseDoesNotExist: 3 ms

✓ getEmptyAssignmentsReport_ShouldReturnEmptyList: 2 ms

> ✓ createCourse_ShouldCreateNewCourse: 19 ms

✓ getAllCourses_ShouldReturnEmptyListIfNoCoursesExist: 1 ms

✓ getLastNameInitialDistribution_ShouldReturnDistribution: 2 ms

✓ getCoursesForStudent_ShouldThrowExceptionIfStudentDoesNotExist: 1 ms

✓ getCoursesForStudent_ShouldReturnOnlyPublishedCourses: 1 ms

✓ getCoursesForStudent_ShouldReturnOnlyPublishedCourses: 1 ms

> ✓ getCourseById_ShouldReturnCourse: 2 ms

✓ getActiveCourses_ShouldReturnOnlyPublishedCourses: 1 ms

✓ getCoursesByMonthReport_ShouldReturnReport: 1 ms

C:\Users\natali\.jdk\jbr-17.0.12-1\bin\java.exe ...

OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended

Process finished with exit code 0

Переписала тест, теперь используется Humcrest:

The screenshot shows an IDE with a project structure on the left and a code editor on the right. The project structure includes folders like `repository`, `service`, `utils`, `CoursesApplication`, `resources`, `test`, `java`, `ru.natali.courses`, `controller`, `it`, `service`, `resources`, `screens`, `application-test.properties`, `course_test_data.csv`, `test-data.sql`, `target`, and `.gitattributes`. The code editor shows the `CourseServiceTest` class with the following code:

```
34 class CourseServiceTest {
164 }
165 @ParameterizedTest
166 @MethodSource("courseProvider")
167 void getCourseById_ShouldReturnCourse_WhenExists(
168     Long id, String name, LocalDate startDate, boolean active) {
169     Course testCourse = new Course();
170     testCourse.setId(id);
171     testCourse.setName(name);
172     testCourse.setStartDate(startDate);
173     testCourse.setActive(active);
174
175     when(courseRepository.findById(id)).thenReturn(Optional.of(testCourse));
176
177     Course retrievedCourse = courseService.getCourseById(id);
178     /*assertEquals(name, retrievedCourse.getName());
179     assertEquals(active, retrievedCourse.isActive());
180     */
181     //можно добавить Hamcrest:
182     assertThat(retrievedCourse, allOf(
183         hasProperty("name", is(name)),
184         hasProperty("active", is(active))
185     ));
186 }
187 }
```

The bottom of the image shows the test execution results. The test `CourseServiceTest` (ru.natali.courses) passed in 692 ms. The test `getCourseById_ShouldReturnCourse` passed in 692 ms. The test results show two parameterized test cases: [1] id=1, name=Java Basics, startDate=2023-01-01, active=true (689 ms) and [2] id=2, name=Python Advanced, startDate=2023-01-01, active=true (3 ms). The process finished with exit code 0.

Tests passed: 2 of 2 tests - 692 ms

C:\Users\natal\jdk\jbr-17.0.12-1\bin\java.exe ...

OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because b

Process finished with exit code 0

Mock-и:

Project

repository

service

utils

CoursesApplication

resources

test

java

ru.natali.courses

controller

LoginControllerTest

it

CourseServiceIntegrationTest

service

CourseArchivingServiceTest

CourseServiceTest

ScheduledArchivingServiceTest

StudentServiceTest

CoursesApplicationTests

resources

screens

application-test.properties

course_test_data.csv

test-data.sql

LoginControllerTest.java

CourseServiceTest.java

CourseSe

23

import static org.mockito.Mockito.*;

24

25

@ExtendWith(MockitoExtension.class)

26

public class StudentServiceTest {

27

28

@Mock 13 usages

29

private StudentRepository studentRepository;

30

31

@Mock 2 usages

32

private PasswordEncoder passwordEncoder;

33

34

@InjectMocks 10 usages

35

private StudentService studentService;

36

37

private Student testStudent; 19 usages

38

private Course testCourse; 19 usages

39

private Review testReview; 9 usages

40

41

@BeforeEach

42

void setUp() {

43

// Инициализация Course с корректными данными

44

testCourse = new Course();

45

testCourse.setId(1L);

Run

StudentServiceTest

StudentServiceTest (ru.natali.cours 826 ms)

Tests passed: 10 of 10 tests - 826 ms

findStudentsByCourseAndLastN 664 ms

loadUserByUsername_ShouldThrov 3 ms

updateStudent_ShouldUpdateExist 7 ms

updateStudent_ShouldThrowRunti 5 ms

loadUserByUsername_ShouldRet1 137 ms

checkPassword_ShouldReturnTrue 3 ms

findByUsername_ShouldReturnNull 2 ms

findByUsername_ShouldReturnStu 2 ms

getStudentsByCourse_ShouldRetui 1 ms

createStudent_ShouldSaveStudent 2 ms

C:\Users\natal\jdk\jbr-17.0.12-1\bin\java.exe ...

OpenJDK 64-Bit Server VM warning: Sharing is only supported for boo

18:01:04,835 |-INFO in ch.qos.logback.classic.LoggerContext[default

18:01:04,837 |-INFO in ch.qos.logback.classic.util.ContextInitializ

18:01:04,837 |-INFO in ch.qos.logback.classic.util.ContextInitializ

18:01:04,837 |-INFO in ch.qos.logback.classic.util.ContextInitializ

18:01:04,837 |-INFO in ch.qos.logback.classic.util.ContextInitializ

18:01:04,839 |-INFO in ch.qos.logback.classic.util.ContextInitializ

milliseconds. ExecutionStatus=INVOKE_NEXT_IF_ANY

ru.natali.courses.service

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
CourseService		97 %		75 %	2	24	1	60	0	20	0	1
StudentService		100 %		100 %	0	10	0	23	0	9	0	1
CourseArchivingService		100 %		n/a	0	5	0	17	0	5	0	1
ScheduledArchivingService		100 %		n/a	0	2	0	5	0	2	0	1
Total	6 of 439	98 %	2 of 10	80 %	2	41	1	105	0	36	0	4